



SCSM 2223 - CROSS-PLATFORM APPLICATION DEVELOPMENT

SEMESTER IV, SESSION 2025/2026

PROJECT PROPOSAL

FRONTEND FRAMEWORK: VUE.JS

REACTIVE UI COMPONENTS AND STATE MANAGEMENT

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SECTION 15

MEMBERS:

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1.0 PROJECT SYNOPSIS AND PROBLEM STATEMENT

Project Title: Software Engineering Smart Academic Advisor (AA) Audit System

Problem Statement:

Managing academic progression is a significant challenge for students who must navigate complex course structures while tracking failed or missed subjects. Traditional methods of checking graduation eligibility are manual and prone to human error, often leading to delayed graduations due to missing credit hours. Our system addresses this by implementing a **Vue.js-based Credit Audit Dashboard**.

By comparing a student's uploaded results against a predefined course structure, the application dynamically identifies "Failed" or "Missed" subjects and calculates the "Remaining Credit Hours" in real-time. Using Vue's reactive state management, the system provides an immediate visual breakdown of a student's academic standing. This ensures both the student and their Academic Advisor have a clear, accurate blueprint of the remaining requirements needed to satisfy graduation criteria, thereby reducing administrative oversight and improving academic planning.

Project Objectives:

- To automate the identification of failed/missed subjects using Vue.js conditional logic.
- To provide a dynamic "Credit Progress Bar" that updates reactively based on completed courses.
- To allow Academic Advisors to filter students based on "At Risk" status.

2.0 IMPLEMENTATION EXAMPLES

Scenario: Software Engineering Smart Academic Advisor System

Our project is a Software Engineering Smart Academic Advisor Audit System. It is designed to help students and Academic Advisors review academic progress more easily by comparing the subjects a student has completed with the required course structure. Instead of checking every subject and credit hour manually, the system provides an interactive dashboard that clearly shows completed, failed, and missed subjects.

The Problem

Traditional academic advising usually depends on manually reviewing student results, course structures, and graduation requirements. This can take a lot of time and may lead to mistakes. Students might only discover that they have failed or missed certain subjects late in their studies, which could delay their graduation. Academic Advisors also need a faster and more reliable way to identify students who may be at risk because of failed subjects or incomplete credit hours.

Vue.js Implementation Requirements

To solve this problem, our team will use Vue.js as the frontend framework to develop a reactive and interactive academic audit dashboard.

1. Reusable Components

The system will be organized into reusable Vue components, including:

- StudentProfile.vue
- CourseChecklist.vue
- CreditProgressBar.vue
- FailedMissedSubjects.vue
- AdvisorFilter.vue
- AuditSummary.vue

Each component will manage a specific section of the dashboard. For example, CreditProgressBar.vue will show the student's credit completion progress, while FailedMissedSubjects.vue will display subjects the student has failed or has not taken yet.

2. Vue Directives

The system will use Vue directives to make the dashboard more dynamic and interactive. Some examples include:

- V-model
- V-if
- V-for
- @click
- :class

v-model will be used for selecting students or filtering subjects.

v-for will be used to display subject lists.

v-if will be used to show warnings such as “At Risk” or “Graduation Not Eligible.”

:class will be used to apply different colors for passed, failed, and missed subjects.

3. Reactive State and Real-Time Updates

One of the main Vue.js features used in this project is reactive state management. When a student's result data is entered or uploaded, the dashboard will automatically update the student's academic status.

The system will calculate:

- Completed Credit Hours
- Remaining Credit Hours
- Failed Subjects
- Missed Subjects
- Graduation Eligibility Status
- At Risk Status

For example, if a student has completed 90 out of 120 credit hours, the progress bar will update instantly and show that 30 credit hours are still remaining.

4. Advisor Filtering Feature

The Academic Advisor will be able to filter students based on their academic status, such as:

All Students

At Risk Students

Students with Failed Subjects

Students with Missing Credit Hours

Students Eligible for Graduation

This feature helps advisors quickly find students who need academic support or guidance.

Student Task

The student's task is to build a Vue.js-based dashboard that allows users to view academic progress in real time. The dashboard should use reusable Vue components, Vue directives, and reactive state to display credit progress, failed or missed subjects, and academic risk status.

3.0 TECHNICAL IMPLEMENTATION PLAN

Architecture:

A Client-Server flow. The Vue frontend will hold the "Master Course Structure" (list of all required subjects). When a student's data is fetched from the Laragon/MySQL backend, the frontend will "diff" the two lists to find missing entries.

Data Strategy:

- **Vue Computed Properties:** You will use these to calculate "Remaining Credits." If $\text{Total_Required} = 120$ and $\text{Completed} = 80$, the UI will instantly show 40 without a page refresh.
- **Conditional Rendering (v-if / v-else):** Use these to highlight failed subjects in red and missed subjects in yellow.

State Management (Pinia): Use a centralized store to keep track of the course structure so that multiple components (the Audit page, the CGPA calculator, and the Advisor view) all see the same data.

4.0 Project Work Plan (Timeline)

We have mapped out our 15-week schedule to make sure we hit the major milestones for **SCSM2223** while keeping the code quality high and AI-free.

Phase / Date Range	What we are working on	Who's doing it
Phase 0: Planning (16/3-23/3)	Our team plans on what system to develop.	All Members
Phase 1: Research & Setup (23/3 – 15/4)	Brainstorming the credit audit logic and defining the course structure. Amir and Adlan will find out how the Software Engineering Course works. Fadzril will draft the system architecture and ERDs, while Daniel works on the mobile-friendly UI wireframes.	All Members
Phase 2: Core Build (16/4 – 28/5)	Setting up the local Laragon environment. Daniel starts building the Vue components (like the course list), and Fadzril handles the data flow to make sure everything connects.	Fadzril & Daniel
Milestone: 29/5	Interim Progress Check: We need to show the core system working with basic data flow and the initial UI layout.	All Members
Phase 3: Advanced Features (1/6 – 20/6)	This is when Fadzril implements the reactive state management so the "Remaining Credits" update instantly. Amir starts debugging and testing the system for any crashes.	Fadzril & Amir
Phase 4: Security & Audit (21/6 – 30/6)	Fadzril handles the backend security (SQLi and XSS protection). Alan finishes the 4-5 page report and checks it through Turnitin to stay under the 20% limit.	Fadzril & Alan

Phase 5: Final Prep (1/7 – 2/7)	Last minute prep for the defense. Amir records the 5-7 minute walkthrough video. We also double check that every line of code is ours so we can defend it on the spot.	All Members
Milestone: 3/7	Presentation and Demo Day: Final walkthrough and oral defense of our code logic.	All Members

4.1 Our Roles & Who Does What

To keep things efficient, we've split the responsibilities like this:

- **Fadzril (Backend & Security Lead):** I'm taking care of the architecture, the Laragon setup, and the security logic like PDO prepared statements to block SQL injections.
- **Daniel (UI Lead):** He's the one building the reactive Vue components and making sure the dashboard looks good on both web and mobile.
- **Alan (Documentation Lead):** His focus is the technical report. He'll make sure our writing is original and the Turnitin similarity index is low.
- **Amir (Testing & Quality Lead):** He's managing the demo video, debugging the app, and making sure the whole logic is solid enough for us to survive the oral defense.